

CAT AND DOG IMAGE CLASSIFICATION USING MOBILENET AND FLASK API

Project Report
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Abstract

This project presents a deep learning web application that classifies uploaded images as either cats or dogs using the MobileNet pretrained convolutional neural network. A Flask-based web interface enables users to upload an image and receive the predicted class together with a confidence score. The lightweight MobileNet architecture provides fast inference while maintaining good classification performance.

Objectives

- Build an image classification model using the MobileNet pretrained architecture.
- Train the model using a labeled cat and dog image dataset.
- Perform image preprocessing and data augmentation.
- Evaluate the trained model using suitable metrics.
- Develop a Flask web application for image upload and prediction.
- Display the uploaded image, predicted class, and confidence score.

Software and Hardware Requirements

Software	Hardware
Python 3.x	Intel Core i5 or above
TensorFlow / Keras	Minimum 8 GB RAM
Flask	5 GB Free Storage
Pillow, NumPy	Internet Browser
VS Code	

Dataset

The dataset contains labeled images of cats and dogs arranged into training and validation folders. Images are resized to 160×160 pixels before being processed by the model.

Methodology

The workflow includes dataset collection, preprocessing, augmentation, MobileNet model creation, training, validation, model saving, Flask integration, and prediction through a web interface.

System Workflow

1. Upload image.
2. Save image.
3. Preprocess image.
4. Load trained model.
5. Predict Cat/Dog.
6. Display prediction and confidence score.

Results

The trained model successfully classifies uploaded images as cats or dogs and displays the prediction with a confidence score through an interactive Flask interface.

Advantages

Fast prediction, lightweight architecture, good accuracy, user-friendly interface, and low computational requirements.

Limitations

The system is limited to two classes and its performance depends on image quality and the diversity of the training dataset.

Future Scope

The project can be extended to classify multiple animal species, support webcam prediction, deploy to cloud platforms, and provide mobile application support.

Conclusion

The project demonstrates the practical use of transfer learning with MobileNet for image classification. Integration with Flask creates a complete end-to-end AI application suitable for educational and real-world demonstrations.

References

- TensorFlow Documentation
- Flask Documentation
- MobileNet: Efficient Convolutional Neural Networks for Mobile Vision Applications
- Python Documentation